GIB Math

1. $\boxed{\text{If } Y-X-T, \text{ then } I(T;X) \geq I(T;Y).}$

$Y-X-T$ by definition means $p(t|x,y) = p(t|x)$. Take mutual information $I(T;X,Y)$ (and its basic properties.)

$$I(T;X,Y) = H(T) - H(T|X,Y) = H(T) - H(T|X) = I(T;X)$$

$$I(T;X,Y) = I(T;Y) + I(T;X|Y) \quad \text{equal}$$

$$\therefore I(T;X) \geq I(T;Y)$$

2. $\boxed{\text{Lemma A.1} \quad \forall (A, \Sigma_g) \; \exists (\tilde{A}) \text{ s.t. } \mathcal{L}(\tilde{A}, I) = \mathcal{L}(A, \Sigma_g)}$

$$\Sigma_g = V \Lambda V^T \text{ (diagonalize)}$$
$$= (V\Lambda^{1/2})(\Lambda^{1/2}V^T) = S^T S$$

$$\mathcal{L} = (1-\beta)\ln|A\Sigma_x A^T + \Sigma_g| - \ln|\Sigma_g| + \beta \ln|A\Sigma_{x|y}A^T + \Sigma_g|$$

determinants

factor out

$$|A\Sigma_x A^T + \Sigma_g| = |A\Sigma_x A^T + S^T S| = \cancel{|S^{-T}||S^{-T}A\Sigma_x A^T S^{-1} + I||S|} = |S^T||S^{-T}A\Sigma_x A^T S^{-1} + I||S|$$

$k = |S^T| = |S|$

$$= k^2 |S^{-T} A \Sigma_x A^T S^{-1} + I|$$

$$\boxed{\tilde{A} = S^{-T} A}$$

$$\mathcal{L} = (1-\beta)\left[\ln k^2 + \ln|\tilde{A}\Sigma_x\tilde{A}^T + I|\right] - \ln k^2 + \beta\left[\ln k^2 + \ln|\tilde{A}\Sigma_{x|y}\tilde{A}^T + I|\right]$$

$$\cancel{\ln|\tilde{A}\Sigma_x\tilde{A}^T + I| \ln|I|} = \boxed{(1-\beta)\ln|\tilde{A}\Sigma_x\tilde{A}^T + I| - \underbrace{\ln|I|}_{0} + \beta\ln|\tilde{A}\Sigma_{x|y}\tilde{A}^T + I|}$$

Let $U^{-1} = U^T$, rotation

$$|S^T U^T||US^{-T}A\Sigma_x S^{-1}U^T + I||US| = k^2|\text{ '' }|$$

$\boxed{US^{-T}A = \tilde{A}}$ infinite variations, choice of U.

3. $\boxed{\frac{d}{dA} \ln |A\Sigma_x A^T + \Sigma_g| = (A\Sigma_x A^T + \Sigma_g)^{-1} 2A\Sigma_x}$

Combination of 2 element-wise matrix derivatives

1. $\frac{d}{dA} \ln|A| = A^{-T}$ if symmetric, A^{-1}

2. $\frac{d}{dA} ACA^T = 2AC$

Chain Rule: $\frac{d}{dA} \ln |A\Sigma_x A^T + \Sigma_g| = (A\Sigma_x A^T + \Sigma_g)^{-1} 2A\Sigma_x$ ✓

4a. $\boxed{\text{For } Q_1 Q_2 v_i = \lambda_i v_i, \ Q_1, Q_2 \in PSD, \ \lambda_i \in \mathbb{R}_{\geq 0}}$

$$v_i^* Q_2 Q_1 Q_2 v_i = \lambda_i \underbrace{v_i^* Q_2 v_i}_{\geq 0} \geq 0$$

★ For non-symmetric matrices, positive λ don't imply positive norms, and vice-versa.

★ Rayleigh Quotient: $\lambda_i = \frac{v_i^* Q_2 Q_1 Q_2 v_i}{v_i^* Q_2 v_i}$

If $v_i^* Q_2 v_i > 0$, $\lambda_i \in \mathbb{R}_{\geq 0}$ ✓

If $v_i^* Q_2 v_i = 0$, $v_i \in \text{Null}(Q_2^{1/2})$ or Q_2, $Q_1 Q_2 v_i = 0$, $\lambda_i = 0$ ✓

4b. $\boxed{\text{For } v_i^T \Sigma_{x|y} \Sigma_x^{-1} = \lambda_i v_i^T, \ \lambda_i \in [0,1]}$

From Schur: $\Sigma_{x|y} = \Sigma_x - \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx}$, $\quad S := \Sigma_{xy}\Sigma_y^{-1}\Sigma_{yx} \geq 0$

$$\Sigma_{x|y}\Sigma_x^{-1} = I - S\Sigma_x^{-1}$$

$S, \Sigma_x^{-1} \geq 0$, so $S\Sigma_x^{-1}$ has positive eigenvalues

$\Sigma_{x|y}, \Sigma_x^{-1} \geq 0$, so $\Sigma_{x|y}\Sigma_x^{-1}$ has positive eigenvalues

per 4a.

$v_i^T \Sigma_{x|y}\Sigma_x^{-1} = \lambda_i v_i^T = v_i^T I - v_i^T S \Sigma_x^{-1}$

Characteristic equation for $\Sigma_{x|y}\Sigma_x^{-1}$ is

$$\left|\underbrace{(1-\lambda)}_{\lambda' \to \text{ eigenvalue of } S\Sigma_x^{-1}} I - S\Sigma_x^{-1}\right|$$

$$\lambda(S\Sigma_x^{-1}) = 1 - \underbrace{\lambda(\Sigma_{x|y}\Sigma_x^{-1})}_{\geq 0} \geq 0 \Rightarrow \boxed{\lambda_i \in [0,1]} ✓$$

4c. $\boxed{\text{For } v_i^T \Sigma_{x|y}\Sigma_x^{-1} = \lambda_i v_i^T, \ v_i^T \Sigma_x v_j = r_i \delta_{ij}, \ r_i > 0}$

Any PSD matrix has a root, $Q = S^2$

Let $\Sigma_x^{-1} = S^2$, and $\Sigma_{x|y}\Sigma_x^{-1} = V^{-1}\Lambda V$

$S^T = S$ left eigen, inverse on other side, eigenvector matrix $V \to V\Sigma_{x|y}\Sigma_x^{-1} = \Lambda V$

If we multiply eigenvectors through the PSD matrix $S\Sigma_{x|y}S$, with cancelling,

$$(VS^{-1})S\Sigma_{x|y}S = V\Sigma_{x|y}S$$

Multiply identity

$$V\Sigma_{x|y}S\underbrace{(SS^{-1})}_{I} = V\Sigma_{x|y}\Sigma_x^{-1}S^{-1} = \Lambda(VS^{-1})$$

So $(VS^{-1})S\Sigma_{x|y}S = \Lambda(VS^{-1})$, then VS^{-1} must be the eigenvector matrix for $S\Sigma_{x|y}S$. Since ~~SE~~ $S\Sigma_{x|y}S$ is PSD, VS^{-1} is orthogonal, so $(VS^{-1})(S^{-1}V^T) = V\Sigma_x V^T$ is diagonal. ✓